Stat 220 Midterm

35 questions

Mark your answers on the bubble sheet.

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1. Which statement correctly describes what a 95% confidence interval means?

   (A) There is a 95% probability that the true value lies inside this interval.

   (B) 95% of the observations in the sample fall inside this interval.

   (C) The true value is within the interval unless the sample was unusual.

   (D) Intervals built this way capture the true value 95% of the time.

2. A trial reports $p = 0.03$. A reporter writes that there is a 3% chance the treatment does nothing. What is wrong?

   (A) Nothing, since that is what a p-value measures.

   (B) The figure should have been doubled, since the test was two-sided.

   (C) The p-value assumes nothing is going on and asks how surprising the data is.

   (D) The p-value is about the sample, so it says nothing about any treatment.

3. A team tests 25 subject lines against open rate. One comes back at $p = 0.03$ and the other 24 show nothing. How much should that interest you?

   (A) A good deal, since 0.03 clears the usual cutoff comfortably.

   (B) A good deal, provided the 24 others were properly randomized.

   (C) Not much: with 25 tests, about one false positive is expected.

   (D) Not much, because open rate is too noisy to test this way.

4. A study of 14 patients reports $p = 0.31$ and the authors conclude the treatment does not work. Which two situations would both produce this result?

   (A) The treatment does nothing, or the study was too small to detect what it does.

   (B) The treatment does nothing, or the significance cutoff was set far too low.

   (C) The sample was biased, or the outcome was measured with error.

   (D) The treatment harms patients, or it helps them by a very small amount.

5. An experiment on 3 million sessions finds a 0.2 second difference in load time with $p < 10^{-9}$. The manager calls it a breakthrough. What do you say?

   (A) The p-value is so small the result must be a computational error.

   (B) Nothing, since that p-value settles the question beyond any doubt.

   (C) At this sample size a tiny difference is detectable; ask whether 0.2 seconds matters.

(D) The test is not valid, since the sample is far larger than these formulas assume.

6. Output for a two-sample comparison reads: difference = 1.4, SE = 0.9, $t = 1.56$, $p = 0.12$, 95% CI $[-0.4, 3.2]$. The manager asks how large the effect could plausibly be. Which number answers that?

   (A) The upper end of the interval, 3.2.

   (B) The p-value of 0.12, which measures the strength of the evidence.

   (C) The difference of 1.4, which is the best single estimate.

   (D) The t-statistic of 1.56, which scales the difference.

7. What makes something a probability model rather than a non-probability one?

   (A) It produces predictions that fall between 0 and 1.

   (B) It states a distribution for the outcome, so uncertainty can be derived.

   (C) It was fitted by maximizing a likelihood rather than minimizing an error.

   (D) It requires the predictors to be independent of one another.

8. A colleague asks for the p-value on the most important variable in a random forest. What do you tell them?

   (A) It is in the output, under variable importance.

   (B) It can be obtained by refitting the forest on bootstrap samples.

   (C) A forest has no probability model, so there is no p-value to give.

   (D) Only the top three variables in a forest have p-values.

9. Which pair of models can AIC legitimately compare?

   (A) A linear regression against a gradient boosted tree fitted on the same rows.

   (B) A regression on 600 rows against the same regression on 900 rows.

   (C) A regression with three predictors against one with seven, on the same rows.

   (D) A regression predicting y against one predicting $\log(y)$.

10. Your outcome is whether a customer renewed, yes or no. What does that fact alone rule out?

    (A) Ordinary linear regression, which can predict below 0 and above 1.

    (B) Any model that uses categorical predictors.

    (C) Tree-based methods, which require a numeric outcome.

    (D) Any model that reports a coefficient rather than a probability.

11. The question is which of five store features most affects sales, with a number attached to the answer. Three colleagues propose a linear regression, a random forest, and a single depth-3 tree. Which serves the question?

    (A) The forest, because it will rank the five features by importance.

    (B) The depth-3 tree, because its splits can be read off directly.

    (C) The regression, because only it attaches an interval to the number.

    (D) Any of the three, since all three use the same five features.

12. A model of repair cost on labour hours returns a slope of 84. What does that number mean?

    (A) Each additional labour hour goes with about $84 more cost.

    (B) About 84% of the variation in cost is explained by labour hours.

    (C) Repairs cost about $84 on average.

    (D) A typical prediction is off by about $84.

13. A regression uses region, which has five levels, and the output lists four coefficients. What happened to the fifth?

    (A) It was dropped for being statistically insignificant.

    (B) It was merged into the intercept along with the other four.

    (C) It is the baseline, and the other four are measured against it.

    (D) Its coefficient is exactly zero and so is not printed.

14. Fuel economy on acceleration time gives a slope of $+1.2$. Add engine power and the slope becomes -0.5, both with small p-values. What is the explanation?

    (A) Slow cars tend to be small; the first slope was comparing engine sizes.

    (B) The sign flip shows the sample is too small for either fit to be reliable.

    (C) One of the two fits must have been computed incorrectly.

    (D) Acceleration time has no real relationship with fuel economy.

15. Residuals against fitted values look shapeless, but residuals plotted against the week the job was recorded drift slowly upward across the year. What does that indicate?

    (A) The relationship with the predictors bends and needs a squared term.

    (B) The outcome should be logged, since the drift suggests growth.

    (C) The rows are not independent, so the reported standard errors are wrong.

    (D) Nothing: residuals are expected to drift when sorted by any column.

16. Two fits on the same rows. `y ~ x` gives a slope of 3.1 with SE 0.4. `y ~ x + z` gives 3.0 with SE 1.9. Here x and z correlate at 0.95. What is going on?

    (A) Adding z revealed that the slope on x was biased all along.

    (B) z duplicates x, so the fit cannot tell which deserves the credit.

    (C) The second fit is wrong, since a standard error cannot grow that much.

    (D) The slope barely moved, so the two fits are equally informative.

17. In `price ~ sqft * waterfront`, with `waterfront` coded 0 or 1, the interaction coefficient is +85. What does it mean?

    (A) Waterfront houses cost about $85 more than inland ones.

    (B) A square foot is worth about $85 more on waterfront than inland.

    (C) Waterfront houses cost about 85% more than comparable inland ones.

    (D) The model gains about $85 of accuracy from the interaction.

18. Two regressions on the same rows: model A has $R^2 = 0.62$, model B has $R^2 = 0.64$ and four more predictors. What settles the choice?

    (A) Model B, since a higher R^2 means a better fit to this data.

    (B) Model A, since fewer predictors is always the safer choice.

    (C) The p-values on the four extra predictors in model B.

    (D) AIC, or error on rows neither model was fitted on.

19. What does least squares actually minimize?

    (A) The sum of the vertical distances from the points to the line.

    (B) The largest distance from any point to the line.

    (C) The sum of the perpendicular distances from the points to the line.

    (D) The sum of the squared vertical distances from the points to the line.

20. A customer asks how long their one repair will take. Which interval belongs in the answer?

    (A) The confidence interval for the average repair.

    (B) The prediction interval for a single repair.

    (C) Whichever is narrower, so the quote stays competitive.

    (D) Neither; a quote should be a single number.

21. Why is the error a model reports on the rows it was fitted on too small?

(A) Because those rows were cleaner than rows collected later.

(B) Because the coefficients were chosen to make those residuals small.

(C) Because any sample is smaller than the population it came from.

(D) Because the error is reported before the fitting has converged.

22. Adding one indicator to a model moves AIC from 934 to 916. A colleague reports that the indicator explains 18% more of the outcome. What is wrong?

(A) Nothing, provided the two models were fitted on the same rows.

(B) AIC went down, so the indicator made the model worse.

(C) The change should have been divided by 934 to get a percentage.

(D) AIC is a score for comparing models, not a share of anything.

23. A lasso fit sends the coefficient on `weekend` to exactly zero. What does that establish?

(A) Weekend has no relationship with the outcome.

(B) Weekend did not earn its keep against this penalty, in this sample.

(C) Weekend would also come out insignificant in an ordinary regression.

(D) Weekend is uncorrelated with the other predictors.

24. You cross-validate a model, then use the same folds to tune a cutoff, then report that cross-validated error as your final number. What is wrong?

(A) Nothing, since cross-validation holds out every row exactly once.

(B) The cutoff was chosen using those folds, so the error is optimistic.

(C) The folds should have been re-randomized between the two steps.

(D) Cross-validation cannot be used on a model with a tunable cutoff.

25. For one new case the prediction interval is $[2.1, 9.8]$ and the confidence interval is $[5.4, 6.1]$. The manager says the model is useless because the first is so wide. What do you say?

(A) They are right, since an interval that wide carries no useful information.

(B) The model needs more predictors until the wide interval narrows.

(C) The prediction interval should be recomputed at a lower confidence level.

(D) The two measure different things; the wide one is the honest one for a case.

26. Two models have cross-validated errors of 0.84 and 0.83 days. The first uses 5 predictors, the second 20. Which do you ship?

(A) The 5-predictor model, since the gain is tiny and it costs less to live with.

(B) Neither: a gap that small means the comparison was done wrong.

(C) The 20-predictor model, since its cross-validated error came out lower.

(D) Whichever turns out to have the higher R^2 on the full data.

27. Forward selection picks `est_boxes` first, and then never uses it again once `volume` has entered. What does that tell you?

(A) Boxes have a non-linear relationship with the outcome.

(B) Boxes matter on their own but not in combination with anything.

(C) The search should be rerun backward to confirm the ordering.

(D) The two columns largely carry the same information.

28. A residual plot shows a narrow band on the left widening steadily to the right. What is the first thing to try?

(A) Modeling the log of the outcome, since the spread grows with the prediction.

(B) Dropping the points on the right, which are the ones producing the wide residuals.

(C) Adding predictors, since the unexplained variation is still large.

(D) Nothing: least squares does not assume equal spread.

29. In a model of log(price) on log(weight), the coefficient on log weight is 1.9. What does that say?

(A) An item 10% heavier costs about 19% more.

(B) An item one unit heavier costs about 1.9 units more.

(C) Weight explains about 190% of the variation in price.

(D) An item 10% heavier costs about 1.9% more.

30. Four repairs take 1, 2, 8 and 10 hours. A tree scores a cut by adding the squared distances from each group's own average. What does it score for the cut that puts $\{1, 2\}$ against $\{8, 10\}$?

(A) 0.0, since each group is predicted by its own average.

(B) 2.5

(C) 12.5

(D) 41.0, the total spread around the overall average.

31. As a tree is allowed to go deeper, error on the rows it was fitted on keeps falling while error on held-back rows falls, bottoms out, then rises. Which depth do you choose?

(A) The depth where the held-back error is lowest.

(B) The shallowest, since simpler models generalize more reliably.

(C) The depth where the two curves are closest together.

(D) The deepest, since it fits the data best.

32. A tree grown on repairs needing 0 to 8 parts is asked about a repair needing 15. What comes back?

(A) A prediction extended along the slope it found among the large jobs.

(B) An error, since the value is outside the range it was grown on.

(C) The average of the rows in the leaf holding the largest jobs it saw.

(D) The overall average, since the case matches no leaf.

33. A tree splits on `supplier` inside the many-parts branch and not inside the few-parts branch. What does that suggest adding to the regression?

(A) A squared term in parts needed.

(B) An indicator for jobs that need an unusually large number of parts.

(C) The supplier variable on its own.

(D) An interaction between supplier and parts needed.

34. Your tree predicts better than your regression, and the owner asks how much the new supplier saves per repair. What do you report?

(A) The tree's prediction, since it has the lower held-out error.

(B) Both numbers, and let the owner decide which to believe.

(C) The raw difference in average repair time between the two suppliers.

(D) The regression's coefficient and interval, from a tree-informed model.

35. A model of log(days) gives a supplier coefficient of -0.22. What does that say on the original scale?

(A) Repairs run about 0.22 days shorter with the new supplier.

(B) Repairs run about 20% shorter with the new supplier.

(C) The supplier accounts for about 22% of the variation in repair time.

(D) Repairs run about 22 days shorter on the longest jobs.